



Coimbatore Institute of Technology

Department of Computing

Artificial Intelligence and Machine Learning | MS 402

Project Summary Report

Feature-Bootstrapped Masked TSFM: Theory, Complexity, and Controlled Ablations for Reproducible Time-Series Pretraining

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Code available at: <https://github.com/vishnu-17o7/TSFM>

1. Project Overview

This project develops a reproducible, large-scale time-series foundation model (TSFM) utilizing a masked latent reconstruction objective. The proposed architecture integrates Reversible Instance Normalization (RevIN), convolutional patch tokenization, a bidirectional Transformer encoder, and an MLP reconstruction head. Unlike existing decoder-only models (e.g., TimesFM) that rely on massive internet-scale observational data, this project addresses data scarcity by introducing a fallback mechanism to feature-bootstrap synthesis, leveraging 44 feature tables to generate synthetic temporal data for pretraining. We conducted rigorous complexity analyses and multi-seed ablations across mask ratios and patch lengths to isolate optimal compute-efficient configurations.

2. High-Performance Computing (HPC) Utilization

High-Performance Computing (HPC) resources were pivotal to managing the heavy computational requirements of training large-scale transformer models on extensive time-series datasets. The HPC cluster facilitated:

- **Parallel Multi-Seed Ablations:** Extensive seed-stratified ablations across configurations were executed simultaneously on multiple GPU nodes to quantify uncertainty and prevent misleading conclusions from deterministic single-seed runs.
- **Large-Scale Pretraining:** The compute environment supported deep training horizons (over 2.11M steps, 62 epochs), enabling detailed late-checkpoint auditing and stability analysis.
- **Throughput Optimization:** By utilizing HPC profiling, we empirically validated attention memory scaling ($\mathcal{O}(BN^2d)$) and epoch runtimes. This confirmed that a patch length of $P = 32$ provides the optimal balance of representational capacity and runtime efficiency.

3. Verified Results and Downstream Performance

The project successfully linked pretraining design choices to downstream forecasting performance. Key verified findings include:

- **Configuration Superiority:** Multi-seed evaluations confirmed that patch length $P = 32$ outperforms $P = 16$ and $P = 8$ in both masked reconstruction quality and runtime efficiency under a fixed step budget.
- **Checkpoint Stability:** The optimal masked Mean Squared Error (MSE) of 0.3304 was achieved at step 2.04M, with a minimal 2.04% drift by the final checkpoint at step 2.11M, proving that validation-based checkpoint selection strongly outperforms recency-based selection.
- **Transfer Learning Efficacy:** The pretrained TSFM model transferred robustly to downstream tasks, outperforming from-scratch baselines on 3 out of 4 tested datasets (Metro, Environmental Sensor Telemetry, and ETTh1).
- **State-of-the-Art Benchmark Comparison:** In direct evaluation against Google's TimesFM, our initialized encoder model achieved **Rank 1** overall. It yielded a **33% lower mean MSE**, winning 5 out of 7 standard forecasting benchmark datasets (ETTh1, ETTh2, Electricity, Traffic, and Weather).